the provisions of chapter 5 of this Code. Where there is a mode point

---CHUNK---

SOURCE: MEPC.397(83).pdf
features and characteristics identified in 4.3.9 or 4.4.8, for each engine operational profile, shall be determined in accordance with the provisions of chapter 5 of this Code. Where there is a mode point condition which is the same among the different engine operational profiles, that is not required to be repeated for each test cycle. The required testing may not necessarily be undertaken on the same physical engine;
.2 the specific emission value determined in accordance with 5.12.6.1 for each engine operational profile shall not exceed the applicable limit value as given by regulation 13;
.3 the multiple engine operational profile parent engine specific emission value shall be determined in accordance with 5.12.6.1 from the highest NO x emission rate, qmgasi as per 5.12.5.2, at each mode point across all the engine operational profiles for which the engine is to be certified;
.4 the parent engine test report for each engine operational profile for which the engine is to be certified shall be included in the technical file as required by 2.4.1.5 together with the determination of the multiple engine operational profile parent engi ne specific

---CHUNK---

SOURCE: MEPC.397(83).pdf
which the engine is to be certified shall be included in the technical file as required by 2.4.1.5 together with the determination of the multiple engine operational profile parent engi ne specific emission value;
.5 the multiple engine operational profile parent engine specific emission value shall be entered under 1.9.6 of the Supplement to the EIAPP Certificate; and
.6 Section 2.2.1 of the Supplement to the IAPP Certificate shall be completed to identify which engines installed on a ship are approved to operate with multiple engine operational profiles.
RESOLUTION MEPC.397(83) (adopted on 11 April 2025)
AMENDMENTS TO THE TECHNICAL CODE ON CONTROL OF EMISSION OF NITROGEN OXIDES FROM MARINE DIESEL ENGINES (NOx TECHNICAL CODE 2008)- 13 -
8.3 Use of multiple engine operational profiles
8.3.1 An engine operational profile shall only be used in accordance with the associated conditions as given in the technical file.
8.3.2 The identification reference of the engine operational profile in use shall be recorded as part of the onboard NO x verification procedure together with data that

---CHUNK---

SOURCE: MEPC.397(83).pdf
associated conditions as given in the technical file.
8.3.2 The identification reference of the engine operational profile in use shall be recorded as part of the onboard NO x verification procedure together with data that demonstrates that the conditions attached to the use of that engine operational profile

were being complied with.
8.3.3 On change from one engine operational profile to another, the date and time of the completion of that change shall be recorded as part of the onboard NO x verification procedure for that engine."
Appendix V Parent engine test report and test data
11 The title of appendix V is replaced by the following:
"Parent engine test report, test data, and determination of the highest composite specific emission value"
12 The title of section 1 is replaced by the following:
"Section 1 Parent engine test report
(see 5.10 and 8.2 of the Code)"
13 The title of section 2 is replaced by the following:
"Section 2 Parent engine test data to be included in the technical file, additionally, for marine diesel engines to which 8.1.1.3 applies, the relevant test data for all engine operational profiles for which the engine is certified which

---CHUNK---

SOURCE: MEPC.397(83).pdf
"Section 2 Parent engine test data to be included in the technical file, additionally, for marine diesel engines to which 8.1.1.3 applies, the relevant test data for all engine operational profiles for which the engine is certified which are to be included in the technical file (see 2.4.1.5 and 8.2 of the Code)"
14 A new section 3 is added after the existing section 2 as follows:
"Section 3 Multiple engine operational profile parent engine, determination of the composite specific emission value to be included in the technical file for engines with those multiple engine operational profiles
(see 8.2 of the Code)
Calculation of the highest composite specific emission value in accordance with 8.2.3.3.
Appendix VII Checklist for an engine parameter check method
15 In paragraph 1, at the end of sub -paragraph 1.14.1, "." is replaced with ";", and sub-paragraphs 1.15 to 1.17 are added after the existing sub-paragraph 1.14 as follows:
"1.15 list of identification references of all engine operational profiles available for the engine and associated conditions, if applicable, under which each is to be used (see chapter 8 of the Code);

---CHUNK---

SOURCE: MEPC.397(83).pdf
"1.15 list of identification references of all engine operational profiles available for the engine and associated conditions, if applicable, under which each is to be used (see chapter 8 of the Code);
1.16 list of auxiliary control devices accepted for the engine and the operating conditions under which those devices function;

1.17 the engine power or engine load and speed boundaries within which the engine is certified to operate."
16 A new appendix IX is added as follows:
"Appendix IX Flow chart for engine certification test cycle determination
(refer to 3.2 of the Code)
"

17 A new appendix X is added as follows:
"Appendix X Calculation of not-to-exceed emission limit value within not-to-exceed zone
(refer to chapters 3 and 5 of the Code)
1 This appendix describes the method for determining the not-to-exceed emission limit value, , at any point within the not-to-exceed zone for comparison with a determined point emission value as set out in 3.3 of this Code.

---CHUNK---

SOURCE: MEPC.397(83).pdf
1 This appendix describes the method for determining the not-to-exceed emission limit value, , at any point within the not-to-exceed zone for comparison with a determined point emission value as set out in 3.3 of this Code.
2 Where engine test results are used to determine a point emission value, formula (1) shall be used to generate that value. At that point, the tolerance requirements of 5.9.6.2 of this Code apply:
 =

(1)
where:
 = NOx at the point Mn in g/kWh
 = Power at the point Mn (brake plus auxiliary) in kW
 = Mass flow rate of NOx in g/h see 5.12.5.2 of this Code
  is to be corrected for humidity and temperature consistent with the method used for the engine test from 5.12.4 of this Code.
Test cycle selection flowchart
Engine is a propulsion engine*
Engine operated with fixed pitch propeller
Yes
Yes
No
Yes
No
No
No
E2
E3
Engine is propeller-law operated
Engine is constant speed
E3
Yes D2
C1
* A propulsion engine may additionally perform non-propulsion duties during or separately to propulsion duties. An engine that solely or in part provides athwartships movement of a ship is not a propulsion engine.

---CHUNK---

SOURCE: MEPC.397(83).pdf
E3
Yes D2
C1

* A propulsion engine may additionally perform non-propulsion duties during or separately to propulsion duties. An engine that solely or in part provides athwartships movement of a ship is not a propulsion engine.
RESOLUTION MEPC.397(83) (adopted on 11 April 2025)
AMENDMENTS TO THE TECHNICAL CODE ON CONTROL OF EMISSION
OF NITROGEN OXIDES FROM MARINE DIESEL ENGINES (NOx TECHNICAL CODE 2008)- 15 -
3 Designation of not-to-exceed zone for E2, E3 and D2 test cycles in limit area of the not-to-exceed zone

3.1 The limit area of the not-to-exceed zone for engines certified to the E2 and E3 test cycles is defined by a speed boundary of equal to or greater than 63% and a power boundary of equal to or greater than 25%. The limit area of the not-to-exceed zone for the D2 cycle is defined by a power boundary of equal to or greater than 25%, at the nominal speed of the engine.

3.2 For the E3 and variable-speed application of the E2 cycle certified engines, the applicant is to define, in accordance with 3.3.1 of this Code, the not-to-exceed zone within the limit area of the not-to-exceed zone as wide or as narrow as applicable

---CHUNK---

SOURCE: MEPC.397(83).pdf
the applicant is to define, in accordance with 3.3.1 of this Code, the not-to-exceed zone within the limit area of the not-to-exceed zone as wide or as narrow as applicable for their intended applications of the engine. The applicant -defined not-to-exceed zone shall encompass all normal steady -state speed load combinations within the